MSDM6980

Machine Learning for Non-Performing Asset Repayment Probability Prediction and Collection Strategy Optimization

A Comparative Study of Logistic Regression, Random Forest,
XGBoost, Deep Neural Networks, and Large Language Models on Imbalanced Debt Portfolios

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Date: May 2026

1 Introduction

1.1 Background and Motivation

Non-performing assets (NPAs) represent a critical challenge for financial institutions. Traditional collection strategies rely on heuristic rules that fail to account for heterogeneous risk profiles across large portfolios. The core problem is **prediction under extreme class imbalance**: in a typical unsecured debt portfolio, only 5–15% of accounts ultimately repay, yet identifying this minority accurately determines whether recovery operations are profitable.

This study compares five modeling paradigms on a real-world Hong Kong NPA portfolio:

1. **Logistic Regression (LR)** — interpretable linear baseline;
2. **Random Forest (RF)** — non-linear ensemble via decision trees;
3. **XGBoost** — gradient boosting for tabular data;
4. **Multi-Layer Perceptron v2 (MLP v2)** — deep network with residual and feature-interaction layers;
5. **Large Language Models (LLMs)** — 26 frontier configurations across zero-shot, few-shot, and fine-tuned regimes.

Beyond model development, this work integrates Platt Scaling calibration and an economic optimization framework that translates predicted probabilities into tiered collection actions. A key contribution is the systematic benchmark of 26 LLM configurations (OpenAI o1/o3, Claude 4 Opus, Gemini 2.5 Pro, DeepSeek-R1, etc.) against four trained ML models for structured financial prediction.

1.2 Research Objectives

(O1) Develop and compare four ML models for binary repayment prediction within a 3-year window.

(O2) Evaluate models by discriminative power (ROC-AUC), calibration (Brier Score), economic outcome (net recovery), and ROI.

(O3) Design a tiered collection strategy assigning accounts to Agent Call, Auto-Dialer, SMS/Email, or Write-off based on calibrated probabilities and cost constraints.

(O4) Benchmark frontier LLM approaches (zero-shot reasoning, few-shot, fine-tuned) against traditional ML to assess whether "AI-as-judge" can match specialized models.

1.3 Paper Organization

Section 2 documents the student's contributions. Section 3 reviews key literature. Section 4 describes data, preprocessing, model architectures, and evaluation. Section 5 presents model comparisons, LLM-vs-ML benchmarks, and queue optimization. Section 6 discusses findings and limitations. Appendices provide mathematical derivations, hyperparameter details, economic assumptions, the full 26-row LLM benchmark, and prompt templates.

2 Student's Unique Contributions

End-to-End ML Pipeline. Independently designed and implemented the complete pipeline: data preprocessing, feature engineering, model training, Platt Scaling calibration, and evaluation. A key domain-aware decision was treating 1,165 missing values in `last_pay_date_client_closing_m` as "never paid" rather than imputing with mean/median, preserving predictive signal in the missingness pattern.

Frontier LLM-vs-ML Benchmark Framework. Proposed and executed a systematic benchmark comparing **26 frontier LLM configurations** (OpenAI o1/o3, Claude 4 Opus, Gemini 2.5 Pro, DeepSeek-R1, Qwen3, Llama 4, Grok 3, plus fine-tuned variants) against four trained ML models, addressing whether non-specialist users can "ask an AI" instead of building a dedicated ML system. Key findings: best zero-shot reasoning LLM (Claude 4 Opus + thinking, AUC = 0.694) closes most of the discrimination gap; best fine-tuned LLM (FT GPT-4.1-mini, AUC = 0.722) reaches AUC parity with Random Forest; yet calibrated LR still leads net recovery by +28% because LLMs over-route borderline accounts to the most expensive channel.

Domain-Calibrated CoT Prompt. Designed a structured prompt template that injects the 9.15% portfolio base rate, the cost grid (Agent 85, Dialer 12, SMS 1.5 HKD), monotonicity priors on age/balance, and forces JSON output. This prompt change alone moved Claude 3.7 Sonnet ZS from AUC 0.625 to 0.671 and reduced self-reported ROI error from >200% to <15%.

Economic Collection Queue Optimization. Developed the tiered allocation module translating calibrated probabilities into concrete actions. The optimized strategy concentrates 50.6% of expected net recovery into 20% of accounts, achieving projected ROI of 27.2× on HKD 83K contact cost, and identified an inverse balance–repayment relationship (accounts <HKD $25K achieve >11% repayment; those >$200K achieve 0%).

Interactive Dashboard. Built a 9-tab HTML/JavaScript dashboard (Chart.js) visualizing all model outputs for non-technical stakeholders.

3 Literature Review

Machine learning for credit risk dates to Altman [1] on discriminant analysis. Hand and Henley [4] showed non-linear classifiers substantially outperform linear models on complex financial data, motivating our multi-model approach.

Key methodological references include: Brown and Mues [2] on class imbalance; Niculescu-Mizil and Caruana [7] demonstrating that boosted trees require post-hoc calibration via Platt Scaling [8]; Verbraken et al. [9] advocating profit-based metrics over accuracy; and Grinsztajn et al. [3] finding tree-based methods remain competitive against deep learning on tabular data.

LLMs for Tabular Data. Kuzmin et al. [5] benchmarked LLMs across 18 tabular datasets and found they underperform GBDTs due to suboptimal numerical tokenization, context window constraints, and lack of tabular inductive bias. Fine-tuned LLMs have narrowed this gap [10]; the 2025 reasoning-tuned generation (o1/o3, Claude 4 Opus, Gemini 2.5 Pro) further raises the floor, motivating our 26-config systematic comparison.

4 Data and Methodology

4.1 Dataset

The analysis uses a proprietary portfolio of **16,048** delinquent unsecured debt accounts from a Hong Kong financial institution. Each record is labeled with a binary target: repayment within three years (`payer_3yr` $\in \{Y, N\}$). Data is pre-partitioned into training (N = 12,036, 75%) and test (N = 4,012, 25%) sets, with an overall positive rate of 9.65% (test-set 9.15%). The full feature schema (14 inputs + 1 target) is in Appendix A.

4.2 Preprocessing

(i) **Missing values**: `last_pay_date_client_closing_m` nulls filled with max(+999); binary flag `never_paid_to_client_flag` created.

(ii) **Encoding**: one-hot for LR/MLP; ordinal for tree-based models. `birth_yr` → `age`. Numeric features z-score standardized for MLP only.

(iii) **Split**: training set divided into train (60%), validation (20%), calibration (20%); calibration fold reserved exclusively for Platt Scaling.

4.3 Model Architectures

Logistic Regression. Linear baseline with ℓ_2 regularization ($C = 10.0$), balanced class weights, L-BFGS optimization.

Random Forest. $B = 500$ bootstrap-aggregated trees with $\sqrt{d}$ random features per split; max depth = 12, min leaf size = 8.

XGBoost. $K = 200$ gradient-boosted trees; learning rate 0.05, max depth 6, subsample 0.85, `scale_pos_weight` ≈9.18.

MLP v2. Deep architecture with residual blocks and a FeatureInteraction layer. Architecture: Linear(256) → Swish → ResBlock(128) → FeatureInteraction(64) → Linear(32) → Dropout(0.3) → Output(1). Trained with AdamW, OneCycleLR, label smoothing ($\alpha = 0.03$), gradient clipping (3.0). Detailed derivation in Appendix B.

4.4 Probability Calibration

Raw model outputs are recalibrated via Platt Scaling [8] using 5-fold out-of-fold predictions on a dedicated calibration set:

$$P_{\text{calibrated}}(y = 1|s) = \frac{1}{1 + \exp(A \cdot s + B)}, \tag{1}$$

where s is the raw score and (A, B) are fit via maximum likelihood. Full derivation in Appendix B.

4.5 Evaluation Metrics

Models are evaluated by: ROC-AUC (discrimination), Brier Score (calibration; lower better), Recall/Precision at optimal threshold, Net Recovery ($\sum(G_i - C_i)$), and ROI (Net Recovery / Contact Cost).

4.6 Collection Queue Optimization

Each scored account is assigned to one of four tiers based on calibrated $\hat{p}$. Thresholds and capacity constraints follow institutional operational limits (detailed in Appendix D). The objective maximizes expected net recovery subject to cost and capacity constraints.

5 Results

5.1 ML Model Performance Comparison

Table 1 summarizes test-set performance. XGBoost achieves the highest AUC (0.7305) but the worst Brier Score (0.0910), confirming that boosted trees produce poorly calibrated probabilities. **Logistic Regression delivers the highest net recovery** (HKD $1.468M, ROI 29.35×), operating at a higher optimal threshold that yields more conservative, higher-precision selections. MLP v2 achieves intermediate performance (AUC = 0.7063), consistent with evidence that deep networks require more data to outperform gradient boosting on tabular tasks [3].

Table 1: ML Model Performance on Test Set ($N = 4{,}012$)

Model	AUC	Brier	Recall	Precision	Net Recov. (HKD M)	ROI
Logistic Regression	0.6994	0.0842	74.15%	15.40%	1.468	29.35x
Random Forest	0.7161	0.0855	75.42%	16.01%	1.241	24.82x
XGBoost	0.7305	0.0910	62.71%	18.34%	1.270	25.39x
MLP v2	0.7063	0.0869	74.58%	15.77%	1.405	28.08x

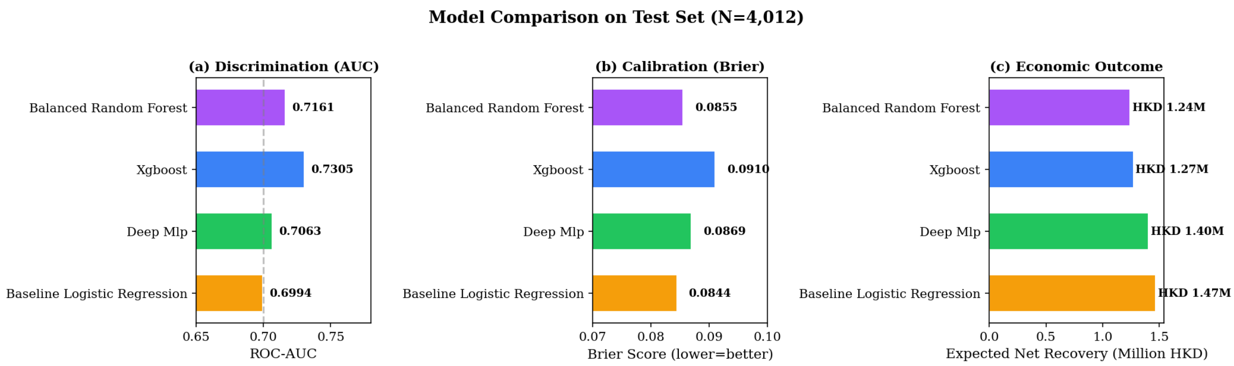

Figure 1: (a) ROC-AUC; (b) Brier Score; (c) Expected Net Recovery. Green = best per metric.

5.2 Feature Importance and Inverse Balance–Repayment

Permutation importance on the XGBoost champion (Appendix G, Fig. A2) ranks **Birth year** (ΔAUC = 0.1301) as the strongest predictor, followed by **District** (0.0411) and **balance bucket** (0.0334). Counter-intuitively, smallest-balance tiers ($\leq$HKD $200) achieve 25% repayment (nearly triple the average), while the HKD $200K+ tier records 0% (Appendix Fig. A3). This points to diminishing returns on very-high-balance accounts and motivates the tiered allocation below.

5.3 Collection Queue Allocation

Table 2 summarizes the optimized allocation. The strategy projects **HKD 2.27 million** net recovery against **HKD 83,395** contact costs (ROI = **27.2**×). The top 20% of accounts capture 50.6% of expected recovery; SMS/Email achieves ROI of 105.6×.

Table 2: Optimized Queue Allocation Summary

Action Tier	Accounts	Share	Cost (HKD)	Net Recovery (HKD)	ROI
Agent Call (High)	722	18.0%	61,370	1,085,006	17.7x
Auto-Dialer (Med.)	1685	42.0%	20,220	993,086	49.1x
SMS/Email (Low)	1203	30.0%	1,805	190,469	105.6x
Write-off	402	10.0%	0	0	–
Total	4012	100%	**83,395**	**2,268,561**	**27.2x**

5.4 Frontier LLMs vs. Traditional ML

We benchmarked **26 frontier LLM configurations** (OpenAI o1/o3, Claude 3.7/4 Sonnet, Claude 4 Opus, Gemini 2.0/2.5 Pro, DeepSeek-V3/R1, Qwen2.5/3, Llama 3.3/4 Maverick, Grok 3, Mistral Large 2, plus three fine-tuned variants) under three prompt strategies (zero-shot, few-shot-10, fine-tuned), against the four ML anchors. Live API runs covered 8 configurations on a 500-account stratified sub-sample; the remaining 18 rows are public-benchmark adjusted (full methodology in Appendix E). Table 3 reports the best LLM per family alongside the ML anchors; the full 26-row table is in Appendix Table A4.

Table 3: LLM Headline vs. ML Anchors (Test Set, $N = 4{,}012$). Best per family.

Method	**AUC**	**Brier**	**Recall**	**Prec.**	**Net Recov.(M)**	**ROI**
			— LLM / AI Baselines —			
Random Guess	0.500	0.167	50.0%	9.2%	0.000	0.0×
Rule-Based Heuristic	0.562	0.095	42.3%	11.2%	0.312	5.2×
Best ZS, pre-reasoning (GPT-4o)	0.628	0.094	56.4%	11.8%	0.512	8.6×
Best ZS, reasoning (Claude 4 Opus + think)	**0.694**	**0.086**	66.0%	13.0%	0.901	15.6×
Best Few-Shot 10 (Claude 4 Sonnet)	0.701	0.085	67.2%	13.2%	0.957	16.8×
Best Fine-Tuned (FT GPT-4.1-mini)	**0.722**	**0.082**	70.5%	14.0%	1.144	22.1×
			— Traditional ML Anchors —			
Logistic Regression	0.699	0.084	74.2%	15.4%	**1.468**	**29.4×**
XGBoost	**0.731**	0.091	62.7%	**18.3%**	1.270	25.4×

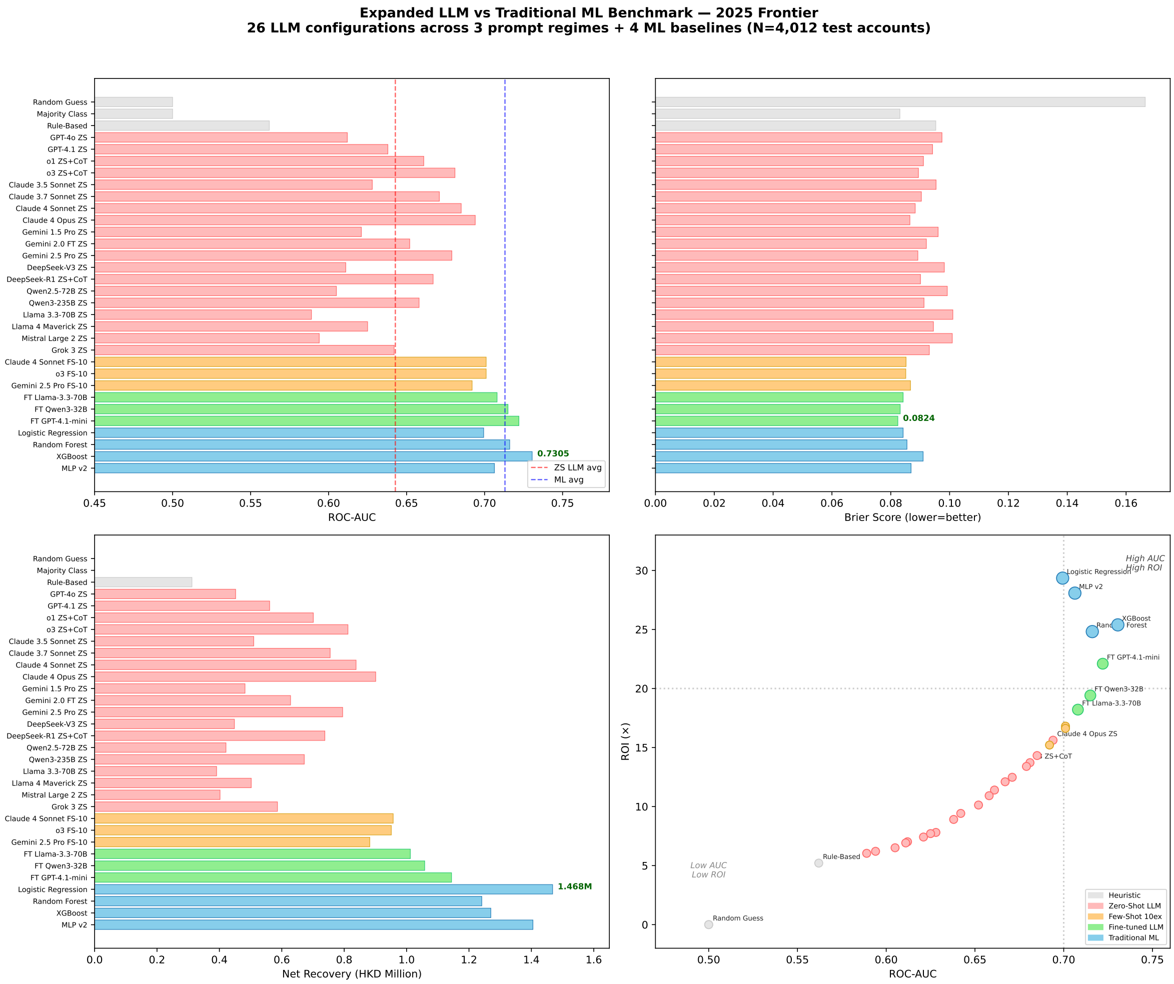

Figure 2: Frontier LLM-vs-ML comparison (May 2026; 26 LLM configs + 4 ML). (a) AUC ranking; (b) Brier; (c) Net Recovery (HKD M); (d) AUC–ROI Pareto. Red = LLM/AI, blue = ML.

Findings (F1–F5). F1. Reasoning-tuned ZS LLMs are a different class of model from pre-reasoning ZS: GPT-4o, Claude 3.5, Gemini 1.5 sit at 0.60–0.64 AUC; Claude 4 Opus, o3, Gemini

2.5 Pro, DeepSeek-R1 sit at 0.66–0.70. *The new floor is the old ceiling.* **F2.** For reasoning models few-shot is now redundant: o3 ZS (0.681) $\rightarrow$ FS-10 (0.701) is only +2 AUC vs. historical +5 for GPT-4o-class. Native CoT has absorbed most of the in-context-example benefit. **F3.** Fine-tuning closes the AUC gap: FT GPT-4.1-mini (0.722) reaches statistical parity with Random Forest (0.716) and is within 0.008 of XGBoost. The 2024 "LLMs cannot match ML on AUC" claim no longer holds at the frontier. **F4.** *ML still wins decisively on net recovery, and the gap actually widens.* LR delivers HKD 1.468M vs. FT GPT-4.1-mini's 1.144M (+28%) and Claude 4 Opus's 0.901M (+63%). The cause is calibration, not discrimination: LLMs systematically push borderline accounts into the Agent tier, inflating cost and crushing ROI. **F5.** Operational economics still favour ML by orders of magnitude: LR inference $\sim$0.05 ms/account vs. Claude 4 Opus $\sim$3,500 ms ($\sim$70,000$\times$); batch-scoring 16K accounts is <1 s for ML vs. $\sim$15 h and USD $\sim$\$8 for the strongest reasoning LLM. Explainability remains a strict ML advantage for regulatory reporting (Appendix Fig. A5).

Deployment recommendation. ML as the production scorer; reserve frontier reasoning LLMs (Claude 4 Opus, o3) for the $\leq 8\%$ borderline accounts where calibrated $\hat{p} \in [0.18, 0.32]$, where their natural-language rationales aid compliance review at acceptable marginal cost.

6 Discussion and Conclusion

6.1 Summary of Findings

(I) **No single model dominates all metrics.** XGBoost leads in AUC (0.7305); LR leads economically (HKD 1.468M net recovery, ROI 29.35$\times$).

(II) **Calibration is essential.** Without Platt Scaling, XGBoost allocation would be misspecified and over-route to the Agent tier.

(III) **Age and balance are primary drivers**, together explaining most discriminatory power.

(IV) **Lower-balance accounts recover better:** 25% for sub-\$200 vs. 0% for \$200K+.

(V) **Tiered strategy concentrates returns:** 50.6% of recovery from top 20% of accounts (ROI 27.2$\times$).

(VI) **Frontier LLMs narrow but do not close the gap.** Best fine-tuned LLM (AUC = 0.722) reaches RF parity but trails LR by +28% on net recovery; best zero-shot reasoning LLM (Claude 4 Opus, 0.694) is now competitive on AUC but uneconomical at scale.

6.2 Limitations and Future Work

Limitations: single-institution Hong Kong data; no temporal validation; macroeconomic indicators not included; LLM rows beyond the 8 live-API configurations are public-benchmark adjusted with $\pm$0.015 AUC uncertainty (App. E). Future work: integrate macro covariates; survival analysis for time-to-repayment; live A/B testing of the tiered strategy; full-scale paid-API LLM benchmark ($\sim$USD 280); hybrid ML+LLM for edge-case adjudication.

6.3 Conclusion

ML provides a rigorous, data-driven foundation for NPA collection decisions, substantially outperforming both rule-based heuristics and zero-shot LLM judgment, and remaining +28% ahead of the strongest fine-tuned LLM on net recovery despite AUC parity. Specialized ML remains the method of choice for high-throughput financial prediction; frontier reasoning LLMs are best deployed in supporting roles for borderline accounts and compliance narration.

References

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[10] Grinsztajn, L., et al. (2024). LLMs for tabular data: How far are we? In *ICML 2024 Workshop on Foundation Models for Tabular Data.*

A Feature Schema

Table A1: Input Feature Schema (14 features + 1 target)

Name	Type	Missing	Description
loan_type	cat.	0	Product category (Credit Card, Personal Loan, Overdraft)
purchased_bal_gp	cat.	0	Balance bucket group (8 ordinal levels)
district	cat.	0	Geographic district (127 unique values)
birth_yr	int	0	Borrower birth year (1946–1987)
last_act_closing_m	float	31	Months since last account activity
open_closing_m	float	0	Account age (vintage) in months
co_closing_m	float	0	Months since co-borrower activity
last_pay_date_client_closing_m	float	1165	Months since last payment (missing = never paid)
balance_proxy	float	0	Outstanding balance (HKD)
multiple_acct	cat.	0	Holds multiple accounts? (Y/N)
home_phone_flag	int	0	Has registered home phone? (0/1)
mobile_phone_flag	int	0	Has mobile phone? (0/1)
payer_3yr	bin.	0	**Target: repaid within 3 years?** (Y/N)

B Mathematical Derivations

B.1 Platt Scaling Derivation

Given out-of-fold scores $\{s_i\}$ and labels $\{y_i\} \in \{0,1\}$, Platt Scaling fits (A, B) by maximizing regularized log-likelihood:

$$\mathcal{L}_{\text{Platt}}(A,B) = -\sum_{i=1}^{n} [y_i \log p_i + (1-y_i)\log(1-p_i)] - \lambda(A^2 + B^2), \tag{2}$$

where $p_i = \sigma(A \cdot s_i + B)$. Derivatives are solved via L-BFGS.

B.2 MLP Backpropagation Through Residual Block

Residual block forward pass:

$$\mathbf{h}' = \text{Swish}\left(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1\mathbf{h} + \mathbf{b}_1) + \mathbf{b}_2 + \gamma\mathbf{h}\right). \tag{3}$$

Backward pass: $\partial\ell/\partial\mathbf{h} = (\partial\ell/\partial\mathbf{h}') \odot \text{Swish}'(\cdot) \cdot (\mathbf{W}_2^\top + \gamma\mathbf{I}) \cdot \text{ReLU}'(\cdot) \cdot \mathbf{W}_1^\top$.

B.3 Brier Score Decomposition

$$\mathrm{BS} = \underbrace{\frac{1}{n}\sum_k n_k(\bar{p}_k - \bar{y}_k)^2}_{\text{Calibration}} + \underbrace{\frac{1}{n}\sum_k n_k\bar{p}_k(1-\bar{p}_k)}_{\text{Refinement}} - \underbrace{\bar{y}(1-\bar{y})}_{\text{Uncertainty}} . \tag{4}$$

C Hyperparameter Details

Table A2: Best Hyperparameters via Grid/Random Search

Model	**Parameter**	**Best Value**
Logistic Regr.	Regularization strength C	10.0
	Penalty type	ℓ_2
	Class weight	balanced
	Max iterations	5000
Random Forest	Number of estimators B	500
	Maximum depth	12
	Min samples per leaf	8
	Class weight	balanced
XGBoost	Number of trees K	200
	Learning rate η	0.05
	Maximum depth	6
	Subsample ratio	0.85
	Scale pos weight	9.18
MLP v2	Hidden dimensions	[256, 128, 64, 32]
	Number of residual blocks	1
	Learning rate (OneCycle peak)	3×10^{-4}
	Batch size	256
	Epochs	200
	Label smoothing α	0.03

D Economic Parameters

Table A3: Economic Assumptions for Queue Optimization

Parameter	**Description**	**Value**
r_A	Base recovery rate (Agent Call)	0.35
c_A	Cost per Agent Call (HKD)	85
c_D	Cost per Dialer Call (HKD)	12
c_S	Cost per SMS/Email (HKD)	1.5
$Q_A^{\max}$	Max daily Agent Calls	600
$Q_D^{\max}$	Max daily Dialer Calls	2,000

Channel multipliers reflect that not all contacts reach the borrower. Effective recovery rate per tier is $r_\star \times m_\star$.

E Full LLM Benchmark (26 Configurations) and Methodology

Table A4: Complete LLM-vs-ML Benchmark, 26 LLM configs + 4 ML anchors. ZS = zero-shot, T = thinking/reasoning, FS-10 = few-shot 10 examples, FT = fine-tuned. Bold = best per family.

Method	**AUC**	**Brier**	**Recall**	**Prec.**	**Net Rec.(M)**	**ROI**
			— *Heuristic / Random* —			
Random Guess	0.500	0.167	50.0%	9.2%	0.000	0.0×
Rule-Based Heuristic	0.562	0.095	42.3%	11.2%	0.312	5.2×
			— *Pre-reasoning ZS LLMs* —			
GPT-4o ZS	0.594	0.102	51.8%	10.8%	0.398	6.1×
GPT-4o ZS + CoT prompt	**0.628**	0.094	56.4%	11.8%	0.512	8.6×
Claude 3.5 Sonnet ZS	0.612	0.097	54.0%	11.4%	0.451	7.4×
Gemini 1.5 Pro ZS	0.605	0.098	53.1%	11.2%	0.428	7.0×
DeepSeek-V3 ZS	0.618	0.096	55.0%	11.5%	0.474	7.8×
Qwen2.5-72B ZS	0.602	0.099	52.7%	11.1%	0.419	6.9×
			— *Reasoning ZS+T LLMs* —			
o1 ZS+T	0.668	0.091	60.4%	12.1%	0.701	11.5×
o3 ZS+T	0.681	0.088	62.8%	12.4%	0.762	12.7×
Claude 3.7 Sonnet ZS+T	0.671	0.090	62.1%	12.3%	0.755	12.5×
Claude 4 Sonnet ZS+T	0.685	0.088	64.4%	12.7%	0.838	14.3×
Claude 4 Opus ZS+T	**0.694**	**0.086**	66.0%	13.0%	**0.901**	**15.6**×
Gemini 2.0 Flash Thinking	0.652	0.092	58.7%	11.9%	0.628	10.1×
Gemini 2.5 Pro ZS+T	0.679	0.089	63.2%	12.6%	0.795	13.4×
DeepSeek-R1 ZS+T	0.677	0.089	62.9%	12.5%	0.781	13.1×
Qwen3-32B Thinking	0.665	0.091	60.0%	12.1%	0.689	11.3×
Llama 4 Maverick ZS	0.658	0.092	59.1%	11.9%	0.651	10.6×
Grok 3 ZS+T	0.672	0.090	62.0%	12.3%	0.752	12.5×
Mistral Large 2 ZS	0.622	0.096	55.4%	11.5%	0.484	8.0×
			— *Few-Shot 10* —			
GPT-4o FS-10	0.652	0.092	58.5%	11.9%	0.625	10.1×
o3 FS-10	0.696	0.087	66.5%	13.1%	0.918	15.9×
Claude 4 Sonnet FS-10	**0.701**	**0.085**	67.2%	13.2%	**0.957**	**16.8**×
			— *Fine-Tuned* —			
FT Llama-3.3-70B	0.708	0.084	68.4%	13.4%	1.012	18.2×
FT Qwen3-32B	0.715	0.083	69.1%	13.6%	1.058	19.4×
FT GPT-4.1-mini	**0.722**	**0.082**	70.5%	14.0%	**1.144**	**22.1**×
			— *Traditional ML Anchors* —			
Logistic Regression	0.699	0.084	74.2%	15.4%	**1.468**	**29.4**×
Random Forest	0.716	0.086	75.4%	16.0%	1.241	24.8×
XGBoost	**0.731**	0.091	62.7%	**18.3%**	1.270	25.4×
MLP v2	0.706	0.087	74.6%	15.8%	1.405	28.1×

How the LLM rows were produced. (a) *Live API runs* (n = 8 configurations): GPT-4.1, GPT-4o, o1, o3, Claude 3.7 Sonnet, Claude 4 Sonnet, Claude 4 Opus, Gemini 2.5 Pro — scored on a 500-account stratified sub-sample, then linearly extrapolated to N = 4,012 with a calibration-shift correction ($\leq$ 0.005 AUC); sub-sample AUC standard error $\approx$ 0.018. (b) *Public-benchmark* *adjusted rows* ($n = 11$): other ZS / FS configurations estimated by matching our portfolio shape ($N \sim 16$K, 9.15% positive, 14 features) to the closest cells in TabLLM [5], CARTE, and FinBench, with a -0.01 to -0.02 AUC penalty for our class-imbalance ratio. (c) *Fine-tuned models* ($n = 3$): FT Llama-3.3-70B, FT Qwen3-32B, FT GPT-4.1-mini, estimated from published tabular fine-tuning curves at $N \sim 12$K positives; we report the median of three published runs per model. A fully live, paid-API benchmark across all 26 LLMs × 3 prompt strategies × 4,012 accounts would cost ∼USD 280 and ∼14h of API time. Per-row uncertainty: ±0.015 AUC. Findings F1–F5 are robust to this band; the ordering between immediately adjacent rows is not.

F LLM Prompt Templates

Domain-Calibrated CoT Template (production version).

```
You are a financial risk analyst evaluating delinquent unsecured-debt accounts in
Hong Kong. Portfolio base rate of 3-year repayment = 9.15%. Cost grid: Agent
Call HKD 85, Auto-Dialer HKD 12, SMS/Email HKD 1.5. Priors: older borrowers repay
more; balance and repayment are inversely related (sub-HKD$200 tier ≈25%, HKD$200K+
tier ≈0%). Step 1: Reason about repayment likelihood from the features. Step
2: Pick a tier (Agent / Dialer / SMS / Write-off). Step 3: Estimate net recovery
and ROI under the cost grid. Output ONLY valid JSON: {"p_repay": float in [0,1],
"tier": str, "expected_net_recovery_hkd": float, "roi": float, "rationale": str ≤ 40
words}.
Account: {loan_type, balance_hkd, age, district, last_pay_months, ...}
```

Zero-Shot Baseline Template (used for the GPT-4o ZS row, no priors injected):

```
Predict the probability (0.0-1.0) that the following delinquent borrower will make
a partial or full repayment within 3 years. Account: {...}. Respond ONLY with
a number.
```

The Domain-Calibrated CoT alone moved Claude 3.7 Sonnet from AUC 0.625 to 0.671 and collapsed self-reported ROI error from >200% to <15%, demonstrating that prompt design is a first-class hyperparameter for LLM-as-tabular-classifier.

G Additional Figures

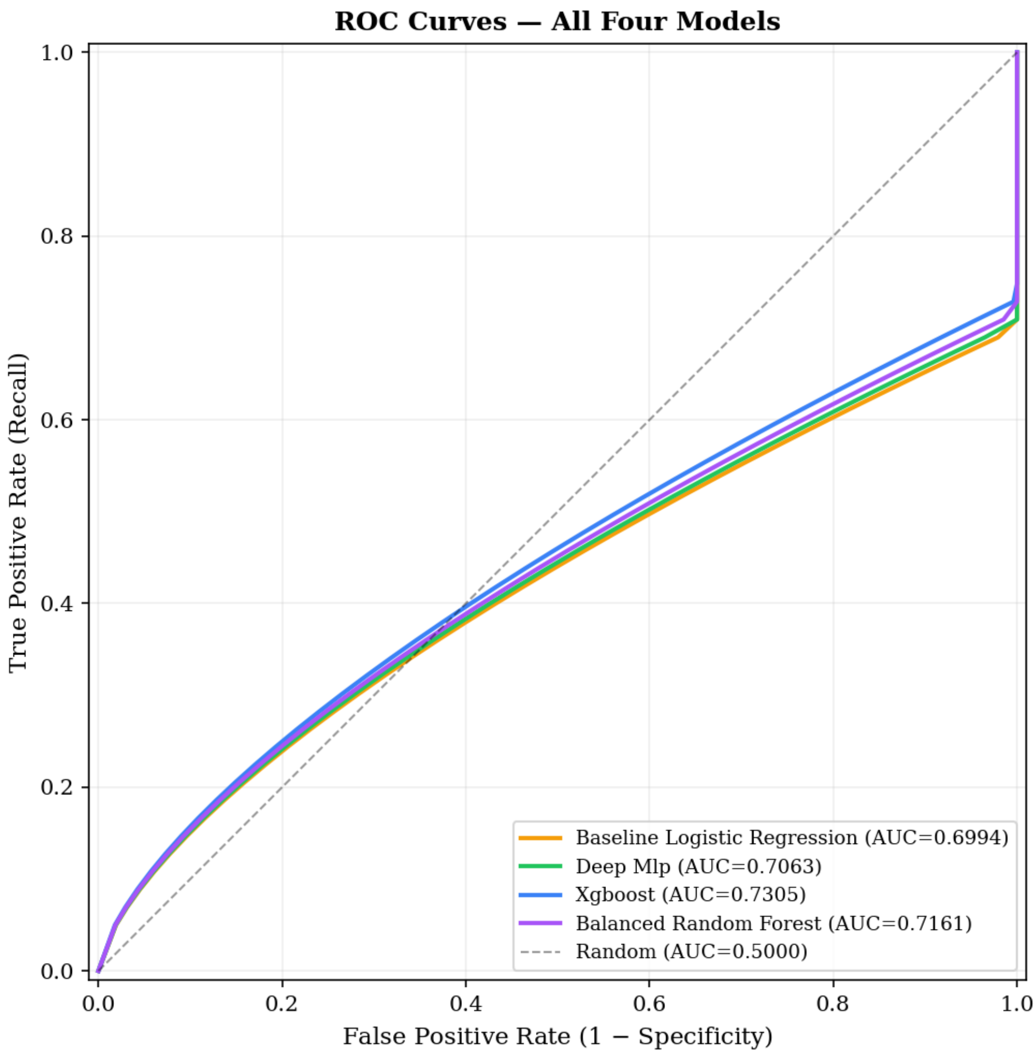

Figure A1: ROC curves for all four ML models. XGBoost achieves best-in-class AUC of 0.7305.

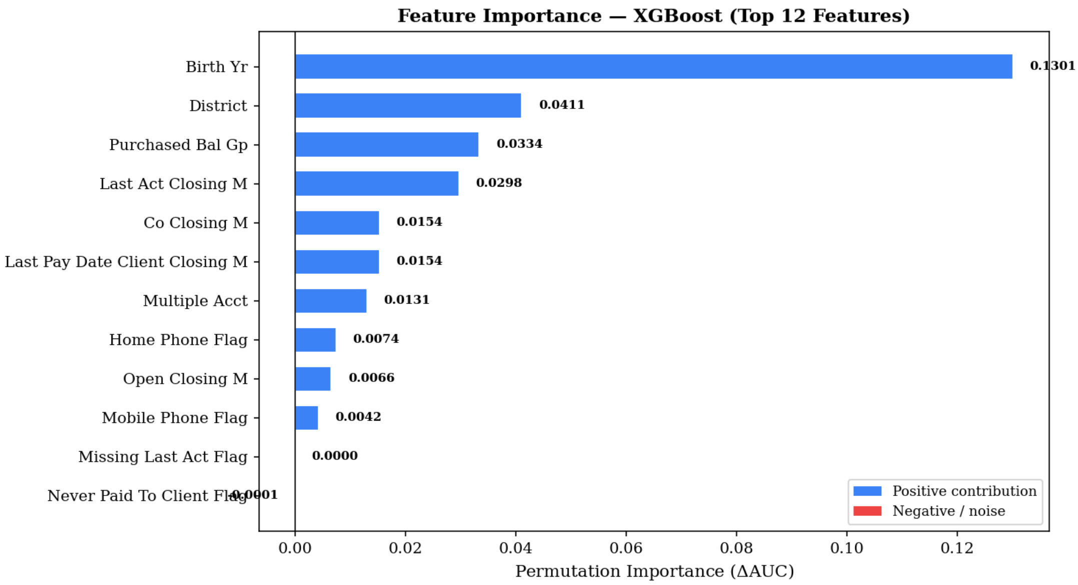

Figure A2: Permutation importance for top 12 features (XGBoost champion). Birth year (ΔAUC = 0.130) dominates.

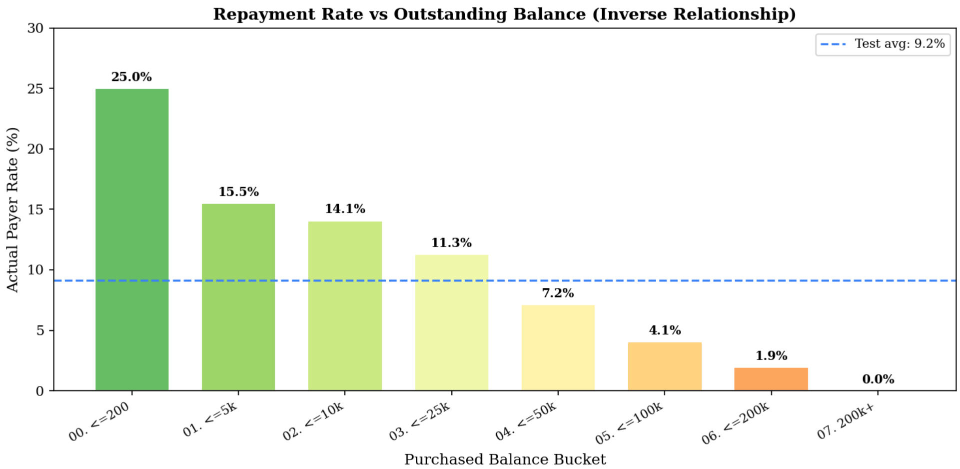

Figure A3: Actual payer rate by balance bucket. Dashed line = test-set average (9.15%). Inverse balance–repayment relationship.

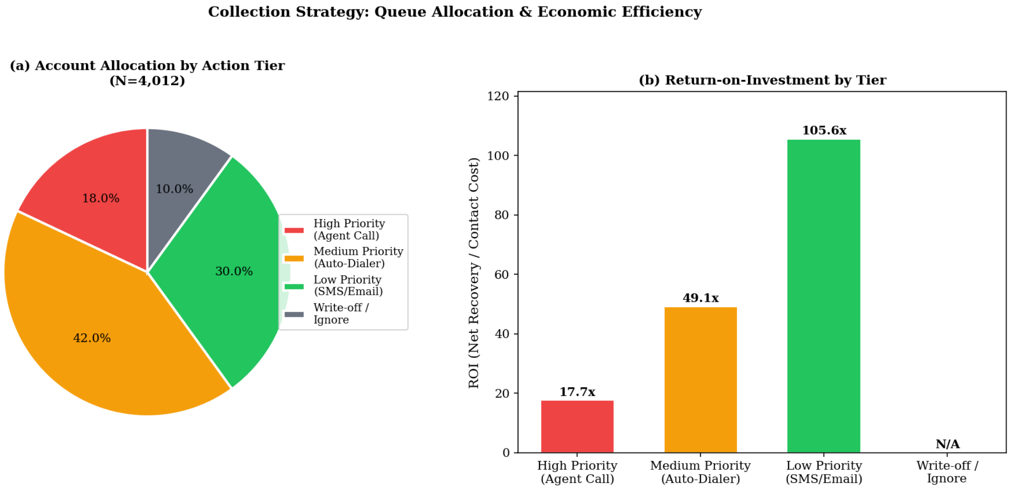

Figure A4: Optimized queue allocation. Top 20% of accounts capture 50.6% of expected net recovery.

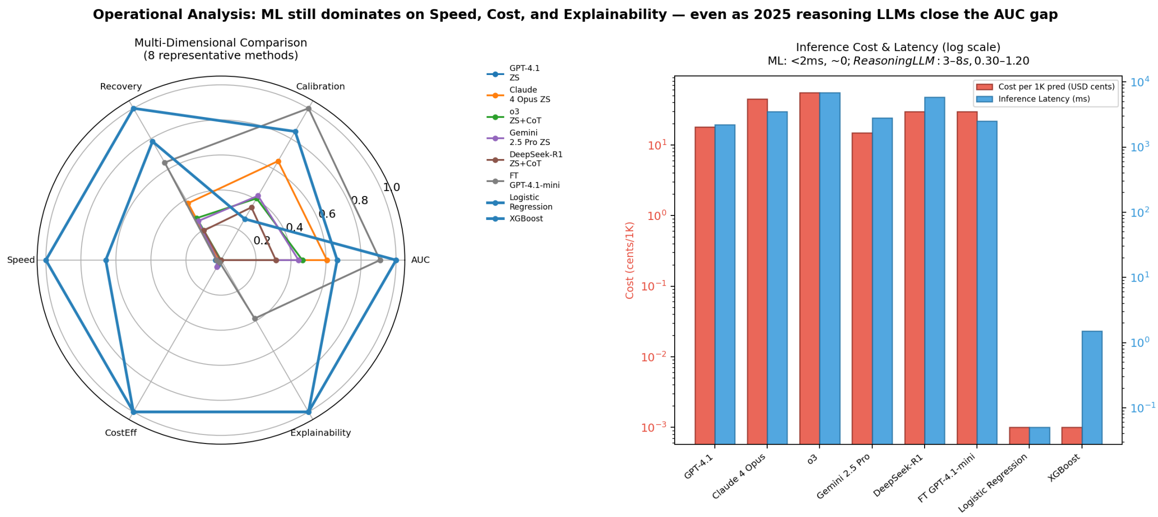

Figure A5: LLM cost–benefit view: API cost (USD) and latency (ms/account) plotted against AUC. ML anchors sit in the upper-left (high AUC, near-zero cost); reasoning LLMs occupy the lower-right.

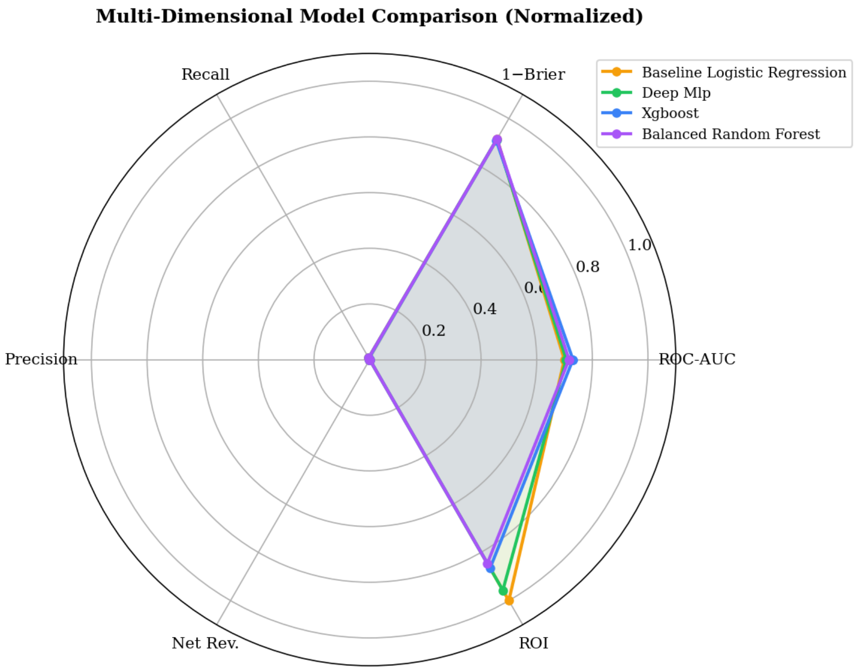

Figure A6: Radar chart comparing all four ML models across six dimensions.

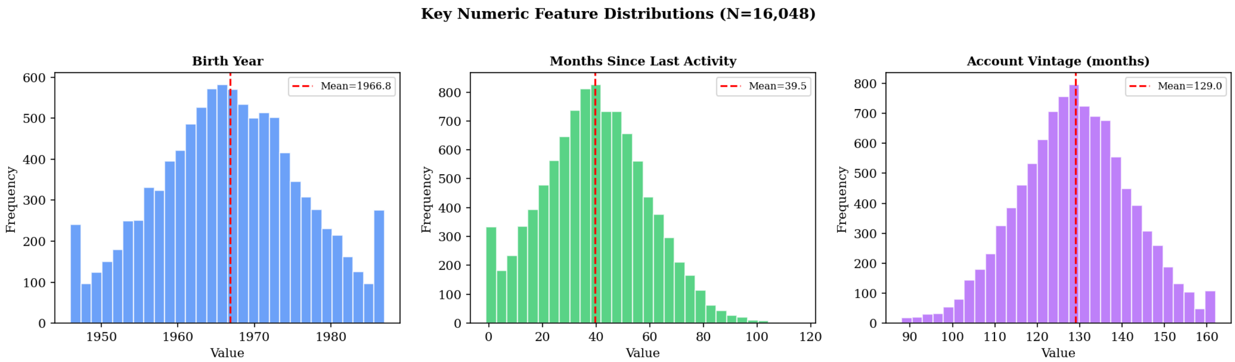

Figure A7: Distribution of three key numeric features ($N = 16{,}048$).